Astronomical Anomaly Detection with Spiking Neural Networks

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Abstract

In modern astronomical surveys, the generated time series data is extensive, and infrequent occurrences are indicative of important phenomena. Identifying anomalies is difficult because they are scarce, complex, and often lack sufficient annotated data for a comprehensive understanding. This project proposes a semi-supervised anomaly detection approach based on spiking neural networks (SNNs) for characterizing normal behavior in temporal signals and identifying anomalies indicative of anomalous events. Astronomical continuous light curves are represented as spike trains to enhance changes and enable the SNN to learn normal behavior. Synthetic time series signal data with anomalies are used for experiments to quantitatively measure performance. To prove the applicability of the proposed approach, it's tested on the available Kepler light curve data. Performance is evaluated based on detection rate, false alarm rate, and anomaly scoring consistency. The project's objective is to evaluate the effectiveness of SNNs for temporal anomaly detection in large-scale astronomical datasets.

1 Introduction

Time-domain astronomy is defined as the study of how astronomical objects change over time, encompassing any naturally occurring physical entities within the universe. From star clusters to galaxies, these systems consist of multiple astronomical bodies or substructures, such as asteroids or planets. The continuously evolving technology involved in astronomy has made the study of time-domain astronomy more feasible and important. Changes that were once too minuscule to observe can now be captured with high temporal resolution. This has permitted researchers to make a step forward in studying rare anomalies, as well as processes related to stellar evolution and planetary dynamics, from events that occur within a span of milliseconds to years [1]. Due to the extremely large temporal datasets in astronomy, this topic presents an opportunity to apply machine learning methods to efficiently identify anomalous patterns.

Anomaly detection plays a key role in astronomical surveys, which often produce datasets of unparalleled scale and greatly increase the probability of scientific discovery. Due to the sheer number and variance of sources, it is impractical to manually inspect or label each piece of data. As a result, traditional supervised learning approaches are infeasible. Methods capable of learning useful representations of astronomical anomalies have thus grown to become increasingly important [2].

Spiking neural networks (SNNs), in particular, provide a basis that mimics biological neural networks. SNNs have been built on artificial neural networks (ANNs) and deep neural networks (DNNs), representing information through discrete spike timings instead of continuous activations [3]. This structure creates an efficient modeling of sparse, time-varying signals that are found in astronomical observations such as light curves and RFI measurements [4]. SNNs have the capacity for efficient, real-time detection of anomalous patterns, which are overlooked by traditional machine learning methods. Since the proposal phase, our focus has been on refining the system design and evaluating feasible approaches prior to implementation. This includes selecting appropriate data sources, defining preprocessing strategies, and determining suitable spike encoding methods for spiking neural networks.

2 Related Work

Recent surveys of SNN research demonstrate their application across multiple domains. In Sabbella et al.'s paper, the authors focus on SNNs for time-series analysis in ubiquitous computing applications, including human activity recognition, speech classification, physiological signal processing, and emotion recognition. The survey highlights several advantages of SNNs for temporal data, where their sparse, event-driven signalling allows for energy-efficient computation, and their spike-based representations naturally capture temporal dynamics more effectively than traditional machine learning models, such as Recurrent Neural Networks, Long Short-Term Memory networks (LSTMs), Convolutional Neural Networks (CNNs), and Transformers. SNNs have also shown strong, real-time processing capabilities, making them a promising approach for resource-constrained and temporal applications [5]. These properties suggest that SNNs are well-suited for analyzing astronomical time-series data, where anomalies may indicate scientifically rare, significant events.

Recent advances in machine learning have shown tremendous progress in the study of astronomical data. For example, Muthukrishna et al. developed RAPID, a real-time automated photometric time-series classification tool that identifies transient phenomena. RAPID employs deep recurrent neural networks with gated recurrent units for classifying astronomical time-series data. RAPID can be applied to ongoing surveys such as the Zwicky Transient Facility (ZTF) and the Large Synoptic Survey Telescope (LSST) [6]. Similarly, within the context of SNNs, Pritchard et al. demonstrate how automated systems can efficiently process dynamic spatio-temporal data through radio frequency interference (RFI) detection. Their study describes the potential of SNNs in time-series functions, achieving competitive performance while discovering meaningful patterns in astronomical datasets [7].

In terms of our project, there has been similar work that follows implementation steps comparable to our methodology. For example, Dennis Bäßler et al. propose an unsupervised anomaly detection method for multivariate time series using SNNs. Their work introduces a rank-order-based learning algorithm that utilizes the precise times of incoming spikes. In addition, their work describes an encoding technique based on multi-dimensional Gaussian receptive fields to handle temporal data. The authors similarly evaluate the effectiveness of their approach using synthetic datasets with various anomaly types, demonstrating strong, effective performance in detecting anomalies in streaming time series data [8].

3 Methodology

3.1 Pipeline Overview

Our methodology has been refined into a structured pipeline consisting of the following stages: data generation, preprocessing, spike encoding, SNN modeling, anomaly scoring, and evaluation. Since the proposal phase, we have evaluated multiple design choices and finalized an approach for each stage of the pipeline. We evaluated both synthetic and real-world astronomical datasets. Synthetic, astronomical-like time-series signals will be used in the initial phase to simulate normal behavior with injected anomalies such as frequency shifts. This provides controlled ground-truth labeling and allows for quantitative evaluation of detection accuracy. To demonstrate real-world applicability, publicly available Kepler light curve data will be incorporated in later stages of development [9].

3.2 Preprocessing

Preprocessing has been identified as a critical step in the pipeline. Time-series signals will be normalized to stabilize input distributions while preserving variations relevant to anomaly detection. We considered multiple spike encoding strategies, including temporal and rate-based encoding, and selected rate encoding for initial implementation due to its simplicity and compatibility with existing SNN frameworks. This encoding converts continuous signals into spike trains, enabling efficient processing by the SNN.

3.3 Training and Evaluation

The SNN model will be trained using only normal data to learn baseline temporal dynamics, following a semi-supervised anomaly detection approach. Anomalies will be identified as deviations from expected spike patterns. Anomaly scores will be computed based on differences between observed and learned behavior. Performance will be evaluated using detection rate, false alarm rate, and anomaly scoring consistency.

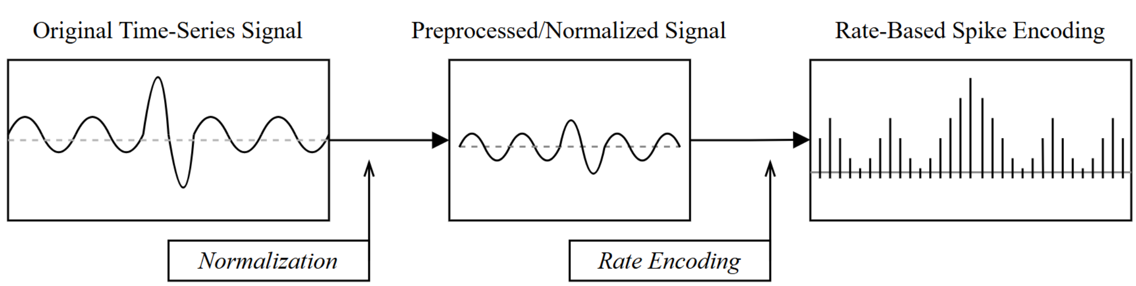

Figure 1: Illustration of the data representation pipeline. Continuous time-series signals are first normalized and then converted into a spike train using rate-based encoding. Signal amplitude will determine spike probability.

4 Intermediate Results

At this stage of the project, we have focused on refining the system design and establishing a clear and feasible pipeline for anomaly detection. While implementation has not yet begun, several key components of the project have been finalized.

Completed Work:

- Defined the overall pipeline architecture for anomaly detection
- Evaluated and selected a data strategy (synthetic data for initial testing, real Kepler data for validation)
- Identified preprocessing techniques such as normalization for stabilizing input data
- Selected rate-based spike encoding as the initial encoding strategy

Observations and Insights: Through this design process, we identified that data representation is a critical challenge when using spiking neural networks, as continuous signals must be converted into spike trains without significant information loss. Additionally, preprocessing plays a key role in ensuring stable and meaningful input for encoding. The lack of labeled anomalies further reinforces the use of a semi-supervised approach.

Challenges:

- Converting continuous time-series data into effective spike representations
- Handling noise and variability in astronomical datasets
- Ensuring stable training of SNN models, particularly with respect to gradient-based learning
- Ensuring that synthetic anomalies are representative of real-world astronomical events

5 Future Work

The next phase of the project will involve implementing and validating each stage of the proposed pipeline in a structured and iterative manner. This will be done by first creating synthetic time series datasets with controlled types of anomalies, such as frequency shifts and amplitude variations, in order to set up a reliable testing environment. The preprocessing techniques, which involve normalization as well as possible noise reduction methods, will also be implemented and validated for effectiveness on signal quality.

Next, the rate-based spike encoding technique will be implemented, and the effectiveness of preserving temporal patterns in the spike trains generated will be analyzed. Then, the SNN model will be constructed and trained using tools such as snnTorch. Initial experiments will be conducted in order to validate the effectiveness of the model in learning normal patterns as well as detecting deviations from expected behavior.

Finally, the performance of the model will be evaluated in terms of performance metrics for anomaly detection, which include accuracy, false positive rate, as well as the consistency of the scores for anomalies. Additionally, the performance of the model with different design choices will be analyzed.

After validation on synthetic data, the model will be used on real-world astronomical datasets, which include the Kepler light curve dataset from NASA, in order to evaluate the performance on real-world data, which includes generalization performance.

Future extensions of this project could involve exploring other advanced spike encoding techniques, as well as other SNN architectures, in order to improve the performance of the model in anomaly detection.

6 References

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